

were in embedded software (82 per cent).

Expansion plans by MNCs and NASSCOM suggest embedded systems to be the next sunrise sector.

Where do I fit?

If you delve a little deeper, you will find that any embedded system is teamwork of three different sets of people: hardware engineers, software engineers and application domain experts. So you can locate your place under any one of these broad categories.

As a hardware engineer, you will be responsible for hardware platform and module prototyping, debugging and testing. Also, you have to ensure compliance with standards and product specifications and initiate design changes as necessary.

If you are comfortable in the software domain, you can start as an embedded systems programmer. This role will engage you in activities like analysis and optimisation of embedded software for the targeted RTOS. Your tasks may include developing installable and built-in device drivers, kernel modifications and embedded application.

Want to explore the application possibilities of the domain? Give a thought to beginning as a wireless protocol engineer. You will be assigned a task for the development, integration and testing of various protocols within an embedded firmware stack used in mobile devices.

Who's hiring?

Almost all industry segments are increasing the use of embedded systems for increased sophistication in the areas of networking, automation and control. As a result, software, telecommunication and electronics companies are heavily diversifying into it. As the application increases, the list of stakeholders can move only one way. The maximum demand is for professionals with 2-5 years of experience. Entry-level hiring is around 20 per cent of the total talent acquisition.

India has always had a pool of talent in the embedded system space. With the complete ownership of the projects on the rise in India, the talent requirement has grown qualitatively as well. Professionals with experience of around five years can command close to Rs 2 million per annum.



—Sunirmal Sen, management consultant, SAP Practice, HCL Technologies

The major recruiters, however, are big players like Texas Instruments, STMicroelectronics, Cadence Design Systems, Honeywell, Delphi, TCS, Wipro, HCL, Infosys, Huawei, Tata Elxsi, Mistral, Poseidon Technologies, Visteon, LG, Portal Player, eInfochips, PaceSoft Silicon, CMC, GDA Technologies, Future Software, ATS, Sankhya Technologies, Bluefont, Intoto, DCM Technologies and Serveen. The list in all probabilities is bound to grow. Further, you should also take into account the SME players in this field. In fact, these are more than 50 per cent stakeholders of the Indian job market.

Do I have it in me?

So now you are seeking the basic criteria for entry, is it? Starting from diploma holders, engineering (as well as science) graduates, postgraduates and even doctorates with the background of electronics/electrical engineering or computer science may try their luck here.

If you consider entry-level recruitment, it is dependent on some predefined skillsets. On the software side, the basic knowledge of 'C' and Assembly language programming is a must. On the other side, sound knowledge of basic electronics, especially digital and analogue design techniques, microcontrollers and microprocessors (e.g., RISC processor or 8051), and system knowledge related to communication protocol and real-time operating system can be helpful in getting a strong

foothold in the field.

If we compare the basic skillsets required by any industry segment in India with the knowledge level of a fresher, a significant drawback of our educational system is revealed. The same holds true for the embedded sector too, which suffers from the lack of quality manpower. In fact, people with knowledge of embedded system design are very rare despite the fact that this is one of the fastest growing sectors of electronics. More so, finding a fresher with a good grasp of the basic electronics and designing concepts is a significant challenge.

What's on 'offer'?

Due to the huge gap between demand and supply of quality human resource, an experienced embedded system engineer may be the most sought after professional in the electronics industry now. In the private sector, the global recession has hugely affected the starting salaries at present. However, analysis of previous year's industry feedback reveals a range between Rs 1.2 million to Rs 2 million per annum for someone with 4-5 years of experience. The compensation depends on the nature of organisation, competency level, academic background of the candidate, etc.

The existing government or public-sector players offer a starting salary in the range of Rs 8000 to Rs 10,000 per month (excluding allowances) for diploma holders and Rs 15,000 to Rs